

CSCI 5832: Natural Language Processing, Fall 2026

Course Information

- **Staff**
 - Instructors: Prof. James H. Martin
 - TAs: Enora Rice and Maria Valentini
- **Time and Location**
 - Tuesday & Thursday 9:30-10:45 in Hellems E210

Natural Language Processing

The field of Natural Language Processing (NLP) is aimed at getting computers to perform useful and interesting tasks involving spoken and written human language. Practical applications of NLP include question answering, machine translation, information extraction, and interactive dialog systems (both written and spoken). Modern NLP systems are based on Large Language Models (LLMs). This class will focus on the foundations of LLMs (probabilistic language models; vector-based representations of meaning; neural networks and gradient-based learning), as well as current model architectures and training recipes.

Course Topics

- Lexicons, Words, and Tokens
- Probabilistic Language Modeling
- Text classification with Logistic Regression
- Vector Semantics and Self-Supervised Learning
- Neural Networks
- Large Language Models
- Transformer Architectures
- Pretraining and Post-Training Approaches

Course Details

Readings: We'll be using draft chapters from the Nth Edition of *Speech and Language Processing* by Dan Jurafsky and James H. Martin. You **should not buy** the current print edition; draft pdfs of the chapters from the new edition are available from the textbook website.

Canvas: We'll be using Canvas for our learning platform. Readings, assignments, and general course information will be posted there. Homework submissions and exams will be done through Canvas as well.

Classroom Videos: Each class lecture will be recorded and made available via Canvas. The slides for each lecture will also be posted to canvas. The primary use of these videos is to review the material and for unavoidable absences from class. That said, this is not a remote or asynchronous class. The videos are not a substitute for coming to class.

iClicker: In many classes there will be questions and activities to which you can respond using iClicker. Scoring for iClicker responses will include participation-based and performance-based points; many of the available points can be earned simply by participating, and questions should be easy to answer correctly if you have been following the lecture. At the end of the semester, your *lowest three* iClicker class scores will be dropped (i.e., you get 3 freebies if you miss class for illness, etc.). *You must be present in the classroom to take the iClicker quiz.*

To participate, you'll need an *iClicker Cloud account (with mobile device)*. Please register for an iClicker Cloud Account and come prepared to use it with a mobile device (e.g., smartphone, laptop, tablet).

Assignments: We will have both written and programming assignments. These will be made available through and submitted through Canvas. Programming assignments will be done in Python along with associated packages and libraries.

Shared Task: The final assignment will be done in groups and will involve the use of data and evaluation methods from a standard NLP task to be determined later in the course.

Quizzes: There will be 3 quizzes during the semester. They will consist of true/false, multiple choice, and short computational questions. Dates are on the class calendar on Canvas.

Grading: Grades will be based on a cumulative score as follows:

- Quizzes (45%; equally weighted)
- Shared task (20%)
- Assignments (15%)
- In-Class Clicker responses (15%)
- Engagement (5%)

The quizzes will assess your understanding of the material covered in class and in the readings. Not everything in the readings will be covered in class, so don't rely on lectures alone to prepare for the quizzes.

Engagement points will be based on a combination of active participation in class, office hours discussions with staff, and anything else that indicates an active engagement with the class.

Collaboration: The collaboration policy is straightforward:

- You are encouraged to discuss assignments with your classmates. However, unless specified otherwise the writing assignments and code that you turn in should be written entirely on your own. For written assignments that involve research and analysis, cite all your sources.
- When in doubt, ask: If you have doubts about this policy or would like to discuss specific cases, please ask the instructor.

Generative AI: You may use AI tools on specific assignments in this course, but their use is limited to the tasks specified in each assignment. The final shared task submission will have specific instructions for the use of AI in citing sources, writing the paper and code development.

Campus Syllabus Statements

Honor Code

All students enrolled in a University of Colorado Boulder course are responsible for knowing and adhering to the [Honor Code](#). Violations of the Honor Code may include but are not limited to: plagiarism (including use of paper writing services or technology [such as essay bots]), cheating, fabrication, lying, bribery, threat, unauthorized access to academic materials, clicker fraud, submitting the same or similar work in more than one course without permission from all course instructors involved, and aiding academic dishonesty. Understanding the course's syllabus is a vital part of adhering to the Honor Code.

All incidents of academic misconduct will be reported to Student Conduct & Conflict Resolution: StudentConduct@colorado.edu. Students found responsible for violating the Honor Code will be assigned resolution outcomes from Student Conduct & Conflict Resolution and will be subject to academic sanctions from the faculty member. Visit [Honor Code](#) for more information on the academic integrity policy.

Accommodation for Disabilities, Temporary Medical Conditions, and Medical Isolation

If you qualify for accommodations because of a disability, please submit your accommodation letter from Disability Services to your faculty member in a timely manner so that your needs can be addressed. Disability Services determines accommodations based on documented disabilities in the academic environment. Information on requesting accommodations is located on the [Disability Services website](#). Contact Disability Services at 303-492-8671 or DSinfo@colorado.edu for further assistance. If you have a temporary medical condition, see [Temporary Medical Conditions](#) on the Disability Services website.

If you have a temporary illness, injury or extended medical treatment for which you require an adjustment of deadlines, alert us of this at your earliest opportunity so we can formulate an accommodation.

Accommodation for Religious Obligations

Instructional faculty members must make every reasonable effort to accommodate all students who have conflicts with scheduled exams, assignment deadlines or required attendance due to a religious observance. Whenever possible, students must notify the instructional faculty member at least two weeks in advance of the expected exam, assignment deadline, or attendance conflict to request an accommodation for religious observance. If the start date of the course is less than two weeks before the date of requested accommodation, students must notify the instructional faculty member on the start date of the course. [See the Student Academic Accommodations for Religious Observances Policy](#) for more information.

Student Names and Pronouns

CU Boulder recognizes that students' legal information does not always align with how they identify. If you wish to have a name other than your legal name appear on your instructors' class rosters and in Canvas, or if you wish to choose pronouns to appear on your instructors' class rosters and in Canvas, visit the [Registrar's website](#) for instructions on how to change your personal information in university systems.

Classroom Behavior

Students and faculty are responsible for maintaining an appropriate learning environment in all instructional settings, whether in person, remote, or online. Failure to adhere to such behavioral standards may be subject to discipline. Professional courtesy and sensitivity are especially important with respect to individuals and topics dealing with race, color, national origin, sex, pregnancy, age, disability, creed, religion, sexual orientation, gender identity, gender expression, veteran status, marital status, political affiliation, or political philosophy.

Additional classroom behavior information

- [Student Classroom and Course-Related Behavior Policy.](#)
- [Student Code of Conduct.](#)
- [Office of Institutional Equity and Compliance.](#)

Sexual Misconduct, Discrimination, Harassment and/or Related Retaliation

CU Boulder is committed to fostering a productive and welcoming learning, working, and living environment. University policy prohibits [protected-class](#) discrimination and harassment, sexual misconduct (harassment, exploitation, and assault), intimate partner abuse (dating or domestic violence), stalking, and related retaliation by or against members of our community on- or off-campus. Denial of an approved accommodation for disability, religious observance, or pregnancy or pregnancy related medical conditions may be discriminatory.

The [Office of Institutional Equity and Compliance \(OIEC\)](#) addresses these concerns, and individuals who have been subjected to misconduct can contact OIEC at 303-492-2127 or email OIEC@colorado.edu. Information about university policies, [OIEC reporting options](#), and [OIEC support resources](#) including confidential services can be found on the [OIEC website](#).

Faculty and graduate instructors are required to inform OIEC when someone discloses misconduct regardless of when or where it occurred. This is to ensure that those impacted receive outreach from OIEC about resolution options and support resources. To learn more about reporting and support options for a variety of concerns, visit the [Don't Ignore It page](#).

Mental Health and Wellness

The University of Colorado Boulder is committed to supporting students' mental health and overall wellbeing. If personal, academic, or emotional challenges are affecting your wellbeing or success, [Counseling and Psychiatric Services \(CAPS\)](#), is here to help. CAPS offers counseling, referrals, psychiatric care, crisis support, and much more. Visit CAPS in the C4C, or call (303) 492-2277, 24/7.

Acceptable Use of AI in This Class

You may use AI tools on specific assignments in this course, but their use is limited to the tasks specified in each assignment. The final shared task submission will have specific instructions for the use of AI in citing sources, writing the paper and code development.